TÉCNICO LISBOA

Thesis Title

Francisco Miguel Carneiro de Castro

Thesis to obtain the Master of Science Degree in

Aerospace Engineering

Supervisor(s): Prof. Full Name 1
Dr. Full Name 2

Examination Committee

Chairperson: Prof. Full Name
Supervisor: Prof. Full Name 1 (or 2)
Member of the Committee: Prof. Full Name 3

Month Year

Dedicated to someone special...

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

A few words about the university, financial support, research advisor, dissertation readers, faculty or other professors, lab mates, other friends and family...

Resumo

Inserir o resumo em Português aqui com o máximo de 250 palavras e acompanhado de 4 a 6 palavras-chave...

Palavras-chave: palavra-chave1, palavra-chave2,...

Abstract

Insert your abstract here with a maximum of 250 words, followed by 4 to 6 keywords...

Keywords: keyword1, keyword2,...

Contents

List of Tables

List of Figures

Nomenclature

Greek symbols

$\phi,\ \theta,\ \psi$ Roll, pitch, and yaw angles, respectively.

Roman symbols

χ Mobile robot orientation.

$\mathbf{p}$ Mobile robot position.

$\mathbf{q}$ Mobile robot pose.

$\mathbf{R}$ Rotation matrix.

$\mathbf{v}$ Mobile robot velocity.

$\mathcal{B}, \mathcal{W}$ Body and world frames, respectively.

Acronyms

3-D Three Dimensional. 6

COM Center of mass. 8, 12–15, 22

DEM Digital Elevation Models. 5–7, 19

DOF Degree of freedom. 2, 8, 9, 19

EVA Extra-vehicular activity. 1

FMM Fast Marching Method. xiv, 20

GNC Guidance, Navigation and Control. 2

HHM Heterogeneous and Hierarchical Models. 5, 6

ICR Instantaneous center of rotation. 9

ISS International Space Station. 1

MER Mars Exploration Rover. 1

MSL Mars Science Laboratory. 1

NLP Non-linear programming. 22–25

NMPC Nonlinear Model Predictive Control. 7

ODE Open Dynamics Engine. 16

ROS Robot Operating System. 6

SA Semantic Attributes. 5, 6

Chapter 1

Introduction

Space robotics covers all robotic systems which operate off the Earth. Firstly, these systems may be classified according to its operational environment, whether it is planetary or orbital. On a second stage, a space robot is comprised of locomotion and autonomy features depending on its goal [**?**].

Space robot locomotion may take several approaches like manipulation, gripping, roving, drilling, sampling, flying, floating, diving or even climbing. On the other hand, a space robot's autonomy ranges from teleoperation to fully-autonomous operations, while other robots execute tasks on a semi-autonomous operation mode.

In 1981, the first orbital robot, Canadarm, was placed on orbital environment with the goal to move payloads from the Space Shuttle to a given deployment position [1]. Later in 2001, Canadarm 2 was launched in order to assist astronauts during EVA, but also to maintain and build the ISS. The humanoid robot Robonaut 2 was launched to the ISS in 2011 to gain experience while performing assisting or teleoperation tasks, for the sake of upgrading it to a future version in order to assist astronauts during EVA. The Moon sampler Surveyor 3 was the first planetary robot to be launched (1967). From the 90s onwards, Mars has become the main exploratory target for surface robots. The *Sojourner* rover, the twin Spirit and Opportunity MER and the MSL have provided valuable scientific information and have enabled ground teams to design rovers which reach higher velocities and larger lifetimes. Recently, in 2021, the Perseverance rover landed on Mars taking on its belly a small robotic helicopter which accomplished the first powered, controlled flight on the thin atmosphere of Mars [**?**].

Robots which operate without human intervention have a limited lifetime, which places a limit on the returned scientifical profit. This thesis will focus on providing a navigation method which aims at delivering a short and safe path while going through in a timely fashion an semi-unknown location.

1.1 Motivation

A conventional planetary, autonomous rover typically assumes a 6-wheeled articulated configuration proposed by Bekker in the 60s, after comparing several designs including tracked or screw-type vehicles [**? ?**]. This configuration fosters crossing of crevasses, rocks or steps up to three times the radius of the wheel, while generating greater stability, redundancy, continuous wheel contact with the terrain to enforce tractability and better distribution of the vehicle load [**?**].

Although this thesis will not focus on robot design, but rather on delivering a proposal for its GNC system, it is worth mentioning alternative configurations. Additional DOF and active suspensions on a 6-wheeled configuration (for instance the rocker-bogie system [**? ?**]) to counteract obstacles in an energy-efficient manner, is an example extensively discussed in the literature. Screw-type vehicles [**?**] like helical geometries and blades are a useful configuration to generate thrust on unconsolidated or amphibious environments, where wheels cannot react against the terrain in order to generate thrust. Alternative configurations like climbing robots have been on experimental stage for space as well. The use of microspines [**?**] by reacting to passive pulling away from a non-horizontal surface is a recent development on climbing robots research. Repelling robots which rely on a tether as an anchor is a proposed method as a way to perform subsurface exploration [**?**].

As it is stated in Table 1.1, launched planetary, autonomous rovers have achieved higher lifetimes as well as higher traverse speeds and travelled greater distances each mission goes by. Below, only rovers deployed in Mars are shown, but the CNSA has also deployed two rovers on the Moon, - *Yutu and Yutu-2* - of which little information is known.

Rover	**Mass (kg)**	**Lifetime (sol)**	**Distance travelled (km)**	**Max. traverse speed (cm/s)**	**Ref.**
Sojourner	10	83	0.106	0.6	[**? ?**]
Spirit	185	2623	7.73	1.0	[**? ? ?**]
Opportunity	185	5352	45.16	1.0	[**? ? ?**]
Curiosity $*$	899	4068	31.15	2.5	[**? ? ?**]
Perseverance $*$	1025	1021	23.776	4.2	[**?**]
Zhurong $**$	240	347	1.921	n.a.	[**?**]

$*$ Still in operation (as of 16 January 2024)

$**$ Inactive (as of 20 May 2022)

Table 1.1: Planetary, autonomous, wheeled rovers achievements.

Although deployed rovers have achieved greater coverages, these have also faced slippage-hazardous situations [**?**] and are subject to get around uncrossable obstacles or stay away from wind devils. These may damage the rover systems, cause progress delay or determine the end of the mission.

Slippage-related hazards may play an important role on preventing a rover to achieve its proposed goals. Even though some slip is desirable to generate traction, excessive slip will result in loss of traction [**?**] and possibly entrap the rover. These events are directly linked to the type of wheel-soil interaction taking place [**?**]. The nature of this interaction takes into account both the type of wheel (rigid or compliable) and its features as well as the type of soil (firm or deformable) and its properties. Apart from it, other variables weighing into slip are the vertical load acting on the soil by the wheel and sinkage, which will cause a decrease on effective thrust and increase drawbar pull [**? ?**]. In this thesis, only rigid wheels are used. These face higher slips on deformable ground, due to high sinkage and shear displacement, while slip is tremendously reduced in the case of compact soils [**?**]. Due to the impossibility of mimicing deformable ground dynamics on real-time simulation test-bed, only rigid terrains are considered. In Table 1.2, one can find slippage events faced by wheeled, autonomous rovers in planetary environment as well as the teleoperating actions put in place to attempt a reversal from the state the rover was at.

Rover	Date of Event	Terrain, Place	Hazard	Action
Curiosity	May, 2015	Deformable terrain (Mars, Gale Crater, nearMount Sharp)	High-slip events	Turn back the rover before it could become entrapped
Curiosity	February, 2014	Deformable terrain (Mars, Gale Crater, nearMount Sharp)	High-slip events (sinkage up to 0.15 [m] and slippage up to 77%)	The route was manually biased to avoid soft sands
Opportunity	April 6, 2010	High windblown ripple dominated by loose, poorly sorted basaltic sands (Mars, Meridiani Planum)	Opportunity experienced significant slippage (up to 40%)	More power to the wheels and extra slip checks
Spirit	May 1, 2009	Soft soil (Mars, Gusev Crater, Home Plate)	Spirit got entrapped with no possibility of recovery	Several recovery techniques (mission concluded on May 24, 2011)
Opportunity	April 26, 2005	Sand dune (Mars, Purgatory Ripple, Meridiani Planum)	Opportunity got entrapped	Recovery actions (for 5 weeks)
Sojourner	August 21, 1997	Deformable terrain (Mars, Ares Vallis)	The right front wheel was parked on a small rock, perching the left front wheel above the surface	Engineers teleoperated the rover to a safer place

Table 1.2: Slippage-related hazards faced by rovers at planetary environments [**?**].

Very small velocities have been achieved by planetary rovers until this point. Apart from lunar teleoperated/driving vehicles, wheeled, autonomous robots have not experienced a dynamic terramechanics regime. In [**?**], a technical review on strategies to increase the operational velocity is presented. In this thesis, a wheeled, autonomous robot is considered to be on high-speed, if it approaches or surpasses 1 m/s, which is roughly ten times larger than the maximum traverse rover velocities presented in Table 1.1.

The Bekker method [**? ? ?**] is capable of providing a definition of the stress fields developed as a result of the interaction between the terrain and the vehicle wheels. However, this method is only accurate on a quasistatic regime, as it fails to accurately quantify soil behaviour in a dynamic regime. Pope stated that the rate of penetration plays a role on determining the normal stress distribution applied on the wheel [**?**]. Therefore, not only the slip ratio but also the vehicle's absolute speed is a factor, when it comes to determining both its sinkage and rolling resistance under an irregular terrain. In 1971, Pope proved that by increasing the vehicle velocity, it is also possible to reduce rolling resistance and its sinkage [**?**]. In 1991, Grahn proposed a model which takes the Bekker pressure-sinkage relationship as its basis, but extends the formulation to include a dynamic regime (considered by Grahn to happen above 1 cm/s), by relating the rate of penetration of a rigid wheel with the wheel driving speed and slip ratio [**?**]. From Grahn's model, it can be inferred that, for a given load, the maximum sinkage decreases with increasing traveling velocities. Additional details on the mentioned models and empirical evidence from planetary missions which involved high-speed vehicles can be found in [**?**].

Secondly, slippage compensation strategies [**?**] shall be included as a safety layer of a navigation control architecture in order to avoid entrapment or embedding situations. These solutions range from torque-based compensation solutions to velocity-based solutions. Torque-based solutions include maximizing traction while ensuring the soil never fails or limiting the torque control signal such that the desired slip matches the estimated slip. These methods - also called motor-current slippage compensation methods - rely on complex terramechanics models, which require the measure of several parameters in real-time, thus limiting the robot's velocity. Conversely, velocity-based slippage compensation methods do not require estimation of parameters from terramechanics models, though these are expected not to perform well in challenging terrain like slopes or sand ripples. This latter approach may include solutions which aim at resolving the so-called kinematic incompatibility problem [**?**] - "wheel fighting" as result of wheel slip on irregular, off-the-road terrains -, adapt control signals with respect to the estimated slip and limit the velocity and acceleration of wheels although this latter approach may result in non-controllable slip hazards because the robot will slip even with a limited motion domain.

As a result of tackling these issues, it is expected to maximize the operational timeline, by decreasing the time dedicated to go through the planetary environment to reach locations of scientific interest and increase the available time for scientific operations. Extending the covered surface is also a desirable navigational outcome, which may be eased by providing the robot with a prior knowledge of the scouting area [**? ?**]. This thesis will focus on providing a navigational control architecture which addresses the issues in the diagnosis presented previously, by taking safety, fastness and covered area as the main metrics of success.

1.2 Problem Statement

1.3 Project outline

1.4 Objectives

1.5 Contributions

1.6 Thesis Outline

Chapter 2

State of the art

2.1 Path planning algorithms

In this section, a review on the generation of paths for mobile robot tracking is presented. Moreover, a review on the modelling of maps for optimal path planning is also presented, based on the review of Burgard et al. [2].

2.1.1 World modelling

World modelling is a *must* when it comes to optimally compute a path for a mobile robot to follow. Due to that, optimal decisions may be taken by a system at any configuration. World modelling falls into the following categories (figure 2.1): indoor and structured environments, natural terrains or dynamic environments. In this thesis, one is interested in performing a metrical evaluation of a mars-like environment with the abscence of dynamic obstacles, so a brief review of models for natural terrains is presented.

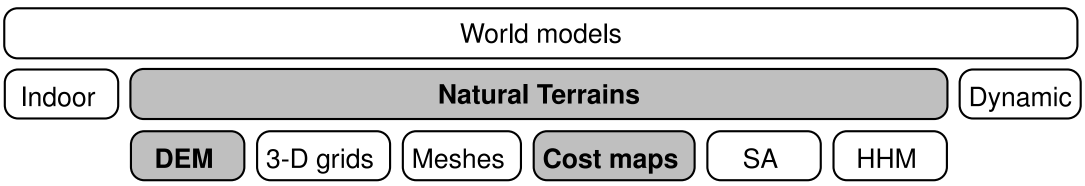


Figure 2.1: World models and map types

Elevation grids or **digital elevation maps** (DEM) are geometric models which store the elevation h of a discrete location (x_i, y_j). Elevation grids are generated due to sensor data. This retrieval of data may be aerial or on the ground. In case of aerial retrieved data, a regularly sampled grid is typically the end result, due to uniform sensor data. If data is retrieved from the ground, the quality of the data depends on the incidence angle and the distance of the sensor to the location (x_i, y_j), which typically yields a variable-sized grid. In the work of Fankhauser et al. [3], a robot-centric, online updated elevation map is estimated, where each cell stores a height estimate $\hat{h}$ and a variance $\hat{\sigma}_h^2$. From a wheeled, mobile planetary exploration point of view, in [4] a robot-centered DEM is built from a 3D point cloud, which was generated by a pair of stereo cameras. Aerial data [5] is also used to deliver an *a-priori* global grid of the target environment, with an accuracy that can be improved by a ground, robot-centered mapping system [4, 6]. Additionally, information on how to tackle issues such as sensor resolution, uncertainty and terrain occlusion can be found in [2].

3-D grids and point sets extend the previous approach to a 3-D representation. Typically, this approach handle large sets of points where free or occupied areas can't be defined directly. Octotrees are a solution to this problem. Then, the Octomap approach [6] uses octotrees to support compact memory representation, multiresolution queries and probabilistic occupancy estimates. A great advantage is that it is possible to model areas which are occluded by vegetation and assign occupancy costs to them.

Meshes present an improvement to Digital Elevation Models and 3-D grids. It is a compact representation and presents continuity over the surface. However, this method requires simplifications techniques to lower computational costs and presents some difficulty on defining complex environments as a continuous surface, such as vegetation. Therefore, precise data is needed to detect discontinuities.

Cost maps are a feasible way to model traversability costs of a rough, uneven terrain and offer a safe and well-known approach to guidance systems on such kind of environments. **Cost maps** may be updated *in-situ* or generated *prior* to placement with overhead imagery. Usually, the computation of costs takes into account the robot kinematics [7] and dynamics [8], as well as terrain inclination. The definition of an explicit cost function is arbitrary, although the robot's limitations are a factor to include. At the end, the obtained traversability map can be combined with a minimum-cost planner [7, 9], in order to output optimal paths into a control system. In the work of Amorim and Ventura [7], a traversability cost is generated at each point by minimizing the height of a mobile robot at each location (x_i, y_j). That minimization is carried out by fixing the center of mass of the robot at (x_i, y_j), and then the mobile robot is rotated until the lowest admissable height is achieved. In the work of Ishigami et al. [9], a criteria function for the **Dijkstra's** algorithm is described, by assigning to each cell a cost, which takes into account the terrain roughness around that cell, distance to goal and terrain inclination. In [10], a **cost map** is created for an occupancy grid, by making obstacle space impossible to traverse and close to obstacle cells progressively less costly to traverse until a safety distance is reached.

Semantic attributes (**SA**) provide information about the type of terrain and objects found. This representation is called a *high-level* one, in the sense that it groups data into broader classes of data, such as traversable or occupied regions, terrain type or vegetation. On the other hand, the previous approaches are defined as *low-level*. Terrain classification methods [11] are used to determine the type of terrain and landmark extraction is useful to detect saliencies in the environment.

Finally, **Heterogeneous and hierarchical models** (**HHM**) offer hybrid maps which combine metric and feature maps. In [12], a `C++` library with a Robot Operating System (ROS) and OpenCV interfaces is described. It manages a multi-layer map for rough terrain navigation. It is designed to store data such as elevation, traversability and surface normals.

2.1.2 Path planning

Chapter 3

Proposed Approach

3.1 Problem formulation

The problem laid out in this thesis is to yield a Nonlinear Model Predictive Control (NMPC) [**? ?**] algorithm $\mathbb{P}$, which models the kinematics and dynamics of a rigid wheeled, autonomous robot $\mathcal{R}$ as well as its interactions with the terrain during the traversal of a semi-unknown, off-the-road, uneven terrain map $\mathcal{M} \subset \mathbb{R}^3$, resembling a planetary environment. It is required that the robot avoids untraversable areas or obstacles $\mathcal{O} \subset \mathcal{M}$, and follows a time optimal traversable pathway to reach a goal point $G \in \mathcal{M}$, starting from a given initial admissable pose $\mathcal{R}_p \in \mathcal{M} \setminus \mathcal{O}$. Moreover, the robot $\mathcal{R}$ shall make a fast, safe and slippage-free traverse, as well as minimizing the energy consumed by the robot battery. The algorithm $\mathbb{P}$ must be computationally optimal and update the system actuation in a timely fashion.

3.2 Optimal path planning

3.2.1 Cost map generation

This thesis is mainly focused on Nonlinear Model Predictive Control, so a simple approach to yield a suitable traversability map is chosen. Both the elevation, $\mathcal{E}$, and traversability, $\mathcal{T}$, maps are assumed to be known *a-priori* and will be computed off-line.

Assumption 1. *The elevation and traversability maps are assumed to be known a-priori without uncertainties, resembling the fact that planetary terrains are scanned prior to the placement of a rover on the ground.*

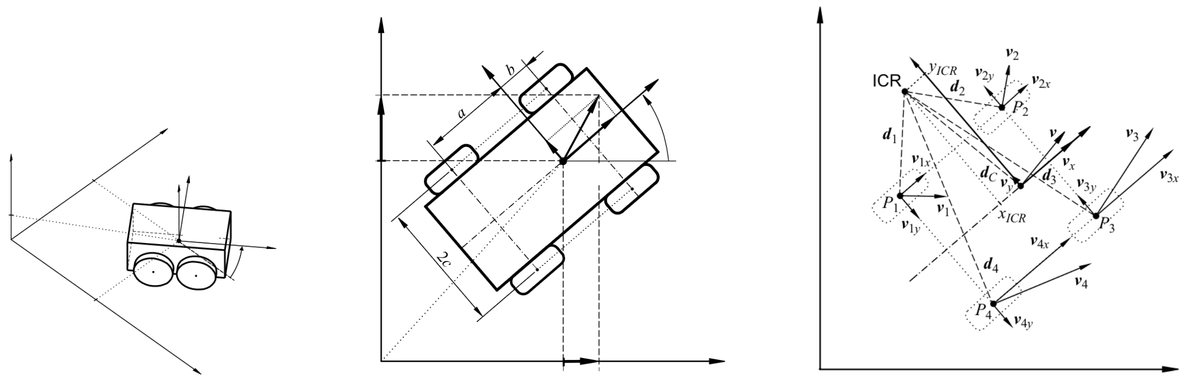
Due to its compactness and simplicity, Digital Elevation Models are chosen to describe the elevation h at each location (x_i, y_j). Then, from the compacted data, a cost map approach is followed in order to yield a traversability map of the environment, by assigining a traversability cost c to each grid point (x_i, y_j). This choice stems from the fact that this is a popular trend in the planetary robotics literature, but also because it is easily combined with a minimum-cost planner algorithm, in order to output a global planner. Finally, the elevation and traversability maps can be joined in a two-layered map $\mathcal{M}_2$, because boths domains are equal. Morever, these frames are considered to share the same frame $\mathcal{W}$, because both map frames match at the origin. An algorithm to address this plan is presented in section 5.1. It is built on top of the works of Amorim and Ventura [7] and Ishigami et al. [9] and provides complete subsection by linking map generation to path planning algorithm and by adding short description of the implementation.

3.3 Mobile robot kinematics

Kinematics describe the motion geometry of a point or body according to the provided actuations. A four-wheeled rover-like kinematics model is to be derived in this chapter, which can be applied both to skid-steering or differentially driven mobile robots.

A wheeled mobile robot $\mathcal{R}$ is moving on a map $\mathcal{M}$, which is centered at the origin of an inertial frame $\mathcal{W}$. A moving frame $\mathcal{B}$ is centered on the robot's center of mass COM and it is fixed to $\mathcal{R}$. Although the mobile robot operates on unstructured, uneven, off-the-road terrains, not all DOF are controllable. This issue is addressed in the following sections.

Assumption 2. *In moving frame $\mathcal{B}$, $\mathcal{B}_x$ points forward, $\mathcal{B}_y$ points to the left and $\mathcal{B}_z$ points upwards. Regarding world frame $\mathcal{W}$, its axis directions are dictated by ROS' standard axis directions [13].*



(a) Robot moving frame (b) Robot geometry (c) ICR

Figure 3.1: Car-like robot kinematics [14]

The mobile robot position and orientation with respect to frame $\mathcal{W}$ is described as $\mathbf{p} = [x\ y\ z]^T$ and $\mathbf{q} = [q_x\ q_y\ q_z\ q_w]^T$, respectively. On the other hand, the mobile robot linear and angular velocities with respect to frame $\mathcal{B}$ stands as $\mathbf{v} = [u\ v\ w]^T$ and $\boldsymbol{\omega} = [p\ q\ r]^T$, respectively. It is decided to use quaternions instead of euler angles to describe the orientation, in order to circumvent ambiguities present on the sum of angles. Moreover, due to the fact that the yaw angle is constrained in $[-\pi, \pi]$, there is a chance that the yaw angle may overcome these limits when the mobile robot is close to them. add section where more information will be given on the issues of euler angles ambiguities. Then, it is possible to map the mobile robot $\mathcal{R}$ linear velocity between frames $\mathcal{B}$ and $\mathcal{W}$ by taking advantage of rotation matrix A.8, but also to map quaternion rates in $\mathcal{W}$ to angular velocities in $\mathcal{B}$ with rotation matrix A.9.

$$\dot{\mathbf{p}} = \mathbf{R}_{\mathcal{B}}^{\mathcal{W}} \mathbf{v} \qquad (3.1) \qquad\qquad \dot{\mathbf{q}} = \bar{\mathbf{R}}_{\mathcal{B}}^{\mathcal{W}} \boldsymbol{\omega} \qquad (3.2)$$

3.3.1 Wheel kinematics

A kinematics model is not only described by its free-body motion equations 3.1 and 3.2, but also by geometrical constraints [14]. Each wheel is actuated by a torque τ_i, which induces on it a spin rate, $\boldsymbol{\omega}_i$, $i = 1, \ldots, N_w$, where $N_w = 4$ for the cases presented in this thesis.

Assumption 3. *The thickness and deformation of each wheel are neglected so that the wheel is assumed to be rigid and to contact the terrain at a single contact point c_i, in case rigid terrain is the working environment [15].*

prove controlability when full model is presented Due to the fact that wheels are fixed, the mobile robot holds two controllable DOF: translation along its longitudinal direction $\mathcal{B}_x$, and rotation around its vertical axis $\mathcal{B}_z$. Therefore, lateral skidding from the vehicle is expected to remain close to zero on flat terrain, although its wheels are prone to skidding when the mobile robot finds itself on uneven or inclined surfaces (figure 3.4). If the mobile robot follows a straight line path, its wheel velocities are tangent to the current path. Moreover, the velocity vector of each wheel is tangent to the terrain that wheel is touching on.

The terrain profile imposes p and q. On the other hand, the existence of holes or bumps force w to be different from zero. v rises above zero in case of lateral skidding.

If a mobile robot is not following a straight line path, this robot will rotate around an ICR point. Each wheel angular velocity is equal to the mobile robot angular velocity, $\boldsymbol{\omega}$. The vector $\mathbf{d}_i$, which connects the ICR point to each wheel position, is orthogonal to each wheel velocity $\mathbf{v}_i = [u\ v\ w]_i^T$, as well as $\mathbf{d}_C \perp \mathbf{v}$, so the following relations may be derived by taking into account the general expression $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$. Vectors $\mathbf{v}_i$, $\mathbf{d}_i$, $\mathbf{d}_C$ and $\mathbf{r}_i$ are all expressed with respect to frame $\mathcal{B}$.

$$\begin{cases} u = qd_{C,z} - rd_{C,y} & u_i = qd_{i,z} - rd_{i,y} \\ v = rd_{C,x} - pd_{C,z} & v_i = rd_{i,x} - pd_{i,z} \\ w = pd_{C,y} - qd_{C,x} & w_i = pd_{i,y} - qd_{i,x} \end{cases} \tag{3.3}$$

Then, it is possible to relate $\mathbf{d}_C$ to $\mathbf{d}_i$, with $\mathbf{d}_C = \mathbf{d}_i + \mathbf{r}_i$, where $\mathbf{r}_i$ is a vector which connects each wheel center's position to the moving frame origin. If this relationship is introduced into equation 3.3, it comes that,

$$\mathbf{v}_i = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}_i \tag{3.4}$$

From the analysis of the previous equation, it is possible to state that,

$$\begin{cases} u_1 = u_2 \\ u_3 = u_4 \end{cases}, \quad \begin{cases} v_1 = v_4 \\ v_2 = v_3 \end{cases}, \quad \begin{cases} w_3 - w_2 = (r_{3,y} + r_{2,y})p \\ w_4 - w_1 = (r_{4,y} + r_{1,y})p \\ w_2 - w_1 = (r_{2,x} + r_{1,x})q \\ w_3 - w_4 = (r_{3,x} + r_{4,x})q \end{cases} \tag{3.5}$$

3.3.2 Wheel-terrain contact

In this thesis, it is important to retrieve each wheel's contact point with the terrain it is touching on, in order to enable the transfer of forces applied at c_i to frame $\mathcal{B}$ and to describe the velocity of each wheel with respect to a frame $\mathcal{T}_i$ with axes $\mathcal{T}_x^i$ and $\mathcal{T}_y^i$ tangent to the terrain at c_i.

Assumption 4. *Every wheel is touching the terrain at a point c_i.*

Lemma 1. *The velocity of each wheel is tangent to the terrain profile at c_i.*

Assumptions 4 and 1 ensure this analysis remains consistent with the fact that, without a contact point c_i between a wheel and the terrain, there would be no reason to obtain a contact angle or a wheel velocity tangent to the terrain.

At every wheel i, a frame $\mathcal{T}_i$ is attached to the wheel center and follows up the terrain profile slope longitudinally. In accordance with assumption 3, roll effect on the determination of c_i is neglected. The transformation from $\mathcal{T}_i$ to $\mathcal{B}$ is based on a rotation matrix expressed in equation 3.6. Frame $\mathcal{T}^i$ cannot yaw with respect to $\mathcal{B}$ because the wheels do not steer ($\psi_i = 0$) and the lateral component, $\mathcal{T}_{y,\,i}$, matches with the wheel axis of rotation ($\phi_i = 0$).

$$\mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} = \begin{bmatrix} \cos(\theta_i) & 0 & \sin(\theta_i) \\ 0 & 1 & 0 \\ -\sin(\theta_i) & 0 & \cos(\theta_i) \end{bmatrix} \tag{3.6}$$

, where θ_i denotes the pitch component of the contact angle. Finally, it is possible to express $\mathbf{v}_i$ with respect to frame $\mathcal{T}^i$ (equation 3.7). This reasoning is depicted in figure 3.2.

$$\mathbf{v}_i = \mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} \mathbf{v}_i^{\mathcal{T}^i} \tag{3.7}$$

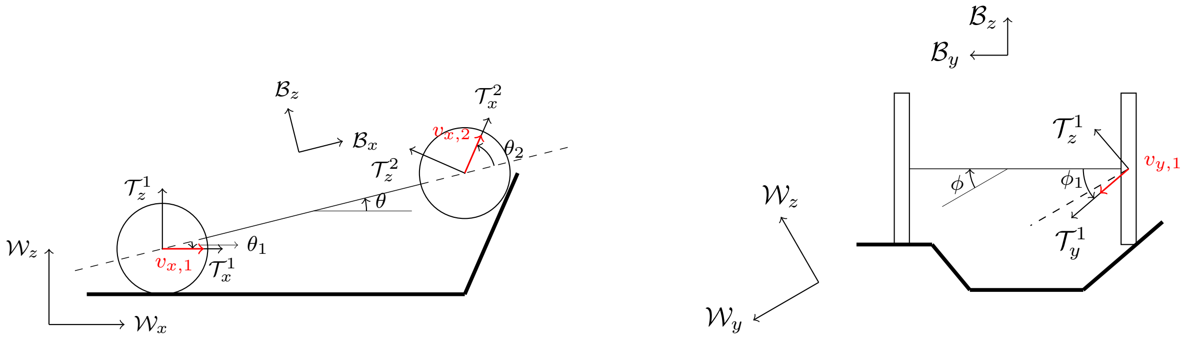
Robot at rest

In this thesis, the state estimation problem is left aside, so the direction of $\mathcal{T}^i$ is obtained by computing a vector $\mathbf{t}$, which is tangent to the surface of the elevation map $e(\cdot)$ at c_i, with coordinates $(x_i,\, y_i,\, e\,(x_i,\, y_i))$, which are ommited for the sake of brevity, from now on.

Definition 1. *The directional derivative of a scalar function $f(x_1,\, \ldots, x_n)$ in the normalized direction $\mathbf{u} = (u_1,\, \ldots, u_n)$, denoted by $\nabla_{\mathbf{u}} f$, is defined to be $\nabla_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.*

If the mobile robot longitudinal, forward bi-dimensional direction vector is $\mathbf{u} = \cos(\psi)\mathbf{i} + \sin(\psi)\mathbf{j}$, from the directional derivative definition (1) it comes that $\nabla_{\mathbf{u}} e = \nabla e \cdot \mathbf{u} = \frac{\partial e}{\partial x}\cos(\psi) + \frac{\partial e}{\partial y}\sin(\psi)$, where $\mathbf{i}$

and $\mathbf{j}$ denote the canonical vectors of $\mathcal{W}_x$ and $\mathcal{W}_y$, respectively. This result expresses the slope of the elevation map $e(\cdot)$ at some point $c_i \in \mathcal{M}_2$ on the direction of $\mathbf{u}$, so $\mathbf{t_u} = [\cos(\psi),\, \sin(\psi),\, \nabla_{\mathbf{u}} e]$.



(a) Longitudinal wheel kinematics. (b) Lateral wheel kinematics.

Figure 3.2: Wheel kinematics

On the other hand, if a leftwards, normalized vector $\mathbf{v} = -\sin(\psi)\mathbf{i} + \cos(\psi)\mathbf{j}$, with $\mathbf{u} \perp \mathbf{v}$, it comes that $\nabla_{\mathbf{v}} e = \nabla e \cdot \mathbf{v} = -\frac{\partial e}{\partial x}\sin(\psi) + \frac{\partial e}{\partial y}\cos(\psi)$. This result expresses the slope of the elevation map $e(\cdot)$ at some point $c_i \in \mathcal{M}_2$ on the direction of $\mathbf{v}$, so $\mathbf{t_v} = [-\sin(\psi),\, \cos(\psi),\, \nabla_{\mathbf{v}} e]$.

Lemma 2. *At some point $(x_i,\, y_i,\, f(x_i,\, y_i)$ of the surface $\mathcal{F}$, the vector, $\mathbf{t_u}$, tangent to $\mathcal{F}$ at the direction of a normalized 2-D vector $\mathbf{u}$, is given $\mathbf{t} = [\mathbf{u}^T, \nabla_{\mathbf{u}} f]^T$.*

Proof. As $\nabla_{\mathbf{u}} f(\mathbf{x}) = \lim_{h\to 0} \frac{f(\mathbf{x}+h\mathbf{u}) - f(\mathbf{x})}{h}$, it is intuitive to state that $\nabla_{\mathbf{u}} f(\mathbf{x})$ provides the slope of $\mathcal{F}$ at some point $\mathbf{x}$ on the direction of $\mathbf{u}$. □

Then, the normalized vectors $\frac{\mathbf{t_u}}{\|\mathbf{t_u}\|}$, $\frac{\mathbf{t_v}}{\|\mathbf{t_v}\|}$ and $\frac{\mathbf{t_u}\times\mathbf{t_v}}{\|\mathbf{t_u}\times\mathbf{t_v}\|}$ may denote the canonical vectors of a frame $\mathcal{T}_i$ with respect to $\mathcal{W}$, which is centered at c_i. From this observation, it comes that,

$$\mathbf{R}_{\mathcal{T}_i}^{\mathcal{W}} = \left[\left(\frac{\mathbf{t_u}}{\|\mathbf{t_u}\|}\right)^T \quad \left(\frac{\mathbf{t_v}}{\|\mathbf{t_v}\|}\right)^T \quad \left(\frac{\mathbf{t_u}\times\mathbf{t_v}}{\|\mathbf{t_u}\times\mathbf{t_v}\|}\right)^T \right] \tag{3.8}$$

Contact points

After obtaining the contact angles, it is possible to deliver each wheel-terrain contact point, c_i, coordinates with respect to $\mathcal{W}$ or $\mathcal{B}$.

At frame $\mathcal{T}_i$, $\mathbf{c}_i = [0,\, r\sin(\phi_i),\, -r\cos(\phi_i)]^T$. Therefore, it comes that,

$$(\mathbf{c}_i)_{\mathcal{W}} = \mathbf{p} + \mathbf{R}_{\mathcal{B}}^{\mathcal{W}} \left(\mathbf{r}_i + \mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}} \mathbf{c}_i\right) \tag{3.9}$$

, where $\mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}} = \left(\mathbf{R}_{\mathcal{B}}^{\mathcal{W}}\right)^T \mathbf{R}_{\mathcal{T}_i}^{\mathcal{W}}$.

The first kinematic constraint (equation 3.10) states that right wheels have the same longidutinal velocity and left wheels also have the same longitudinal velocity, in frame $\mathcal{B}$. The second kinematic constraint (equation 3.11) states that front wheels have the same lateral velocity and back wheels also have the same lateral velocity, in frame $\mathcal{B}$.

$$\begin{cases} (u_1)_{\mathcal{T}} \cos(\theta_1) + (v_1)_{\mathcal{T}} \sin(\theta_1)\sin(\phi_1) = (u_2)_{\mathcal{T}} \cos(\theta_2) + (v_2)_{\mathcal{T}} \sin(\theta_2)\sin(\phi_2) \\ (u_3)_{\mathcal{T}} \cos(\theta_3) + (v_3)_{\mathcal{T}} \sin(\theta_3)\sin(\phi_3) = (u_4)_{\mathcal{T}} \cos(\theta_4) + (v_4)_{\mathcal{T}} \sin(\theta_4)\sin(\phi_4) \end{cases} \tag{3.10}$$

$$\begin{cases} (v_1)^{\mathcal{T}} \cos(\phi_1) = (v_4)^{\mathcal{T}} \cos(\phi_4) \\ (v_2)^{\mathcal{T}} \cos(\phi_2) = (v_3)^{\mathcal{T}} \cos(\phi_3) \end{cases} \tag{3.11}$$

The third constraint states that the difference between vertical velocities of left wheels (or right wheels) in frame $\mathcal{B}$ is equal the vehicle pitch rate times the longitudinal distance between these wheels. Lastly, the fourth constraint states that the difference between vertical velocities of back wheels (or front wheels) in frame $\mathcal{B}$ is equal the vehicle roll rate times the lateral distance between these wheels.

$$\begin{cases} -(u_1)^{\mathcal{T}^1} \sin(\theta_1) + (v_1)^{\mathcal{T}^1} \cos(\theta_1)\sin(\phi_1) + (u_2)^{\mathcal{T}^2} \sin(\theta_2) - (v_2)^{\mathcal{T}^2} \cos(\theta_2)\sin(\phi_2) = (a+b)q \\ -(u_4)^{\mathcal{T}^4} \sin(\theta_4) + (v_4)^{\mathcal{T}^4} \cos(\theta_4)\sin(\phi_4) + (u_3)^{\mathcal{T}^3} \sin(\theta_3) - (v_3)^{\mathcal{T}^3} \cos(\theta_3)\sin(\phi_3) = (a+b)q \end{cases} \tag{3.12}$$

$$\begin{cases} -(u_1)_{\mathcal{T}} \sin(\theta_1) + (v_1)_{\mathcal{T}} \cos(\theta_1)\sin(\phi_1) + (u_4)_{\mathcal{T}} \sin(\theta_4) - (v_4)_{\mathcal{T}} \cos(\theta_4)\sin(\phi_4) = 2cp \\ -(u_2)_{\mathcal{T}} \sin(\theta_2) + (v_2)_{\mathcal{T}} \cos(\theta_2)\sin(\phi_2) + (u_3)_{\mathcal{T}} \sin(\theta_3) - (v_3)_{\mathcal{T}} \cos(\theta_3)\sin(\phi_3) = 2cp \end{cases} \tag{3.13}$$

Provided that p, q, $(u_i)_{\mathcal{T}^i}$ and $(v_i)_{\mathcal{T}^i}$ are known, the contact angles ϕ_i

Simulate robot first without contact angle (assume plane of contacts is parallel to the same plane of the robot; then, include this estimation into the simulation; at rest (for maps generation), use gradients

Don't forget to include the pure rolling condition; it is violated when there is slip

3.3.3 Slippage

During the traversal of a planetary environment, a wheeled mobile robot may experience slip at each wheel i, both on the longitudinal (s_i) and lateral (β_i) directions of the wheel frame $\mathcal{T}_i$, where s_i stands for the slip ratio at wheel i and β_i denotes the slip angle at wheel i. Although the slip angle β_i may be measured at each wheel frame $\mathcal{T}_i$, it is more usual to do it in case of steering wheels. Instead, the slip angle is computed with the mobile robot COM velocities within frame $\mathcal{B}$.

$$s_i = \begin{cases} 1 - \frac{(u_i)_{\mathcal{T}_i}}{r\,\omega_i}, & \text{if} \quad |r\,\omega_i| \geq |\,(u_i)_{\mathcal{T}^i}\,| \text{ (slip)} \\ \frac{r\,\omega_i}{(u_i)_{\mathcal{T}_i}} - 1, & \text{if} \quad |r\,\omega_i| < |\,(u_i)_{\mathcal{T}^i}\,| \text{ (skid)} \end{cases} \qquad \beta = \begin{cases} \arctan\left(\frac{v}{u}\right), & (u \neq 0) \\ \text{sgn}(v) \cdot \frac{\pi}{2} & (u = 0) \end{cases} \tag{3.14}$$

, where r denotes the wheel radius. Physical parameters related to every wheel are equal. If $0 \leq s_i \leq 1$, it is said wheel i is slipping. If $-1 \leq s_i < 0$, it is said that wheel i is skidding or braking. However, if the signs of $(u_i)_{\mathcal{T}^i}$ and $\dot{q}_i$ are opposite to each other, it means $1 < |s_i| \leq 2$. Although these impractical slips are possible [16], no terramechanics analytical models are presented, because no literature was found with research on the topic. Regarding the slip angle β, this quantity is as high as the lateral skid, v, is with respect to the longitudinal velocity, u.

In accordance to the definitions presented above, it is possible to define a new quantity, δ_i, which expresses the relative velocity between wheel i at c_i and the surface.

$$\delta_i = r\omega_i - (u_i)_{\mathcal{T}_i} \tag{3.15}$$

In case $\delta_i = 0$, wheel i presents a pure rolling motion. Apart from that, wheel i exhibits a slipping or skidding behaviour.

Lemma 3. *Pure rolling condition is respected in case equation 3.15 equals zero.*

3.4 Mobile robot dynamics

A dynamics model describes the motion of a point or body subject to external forces. These forces may include gravity, wind blows, wheel-terrain traction or external agents disturbances.

On a mobile robot real-life application, a dynamics model is mainly triggered by gravity and wheel torques, which are actuated by electrical currents. A velocity change is to be expected, which will in turn yield an updated pose (figure 3.3).

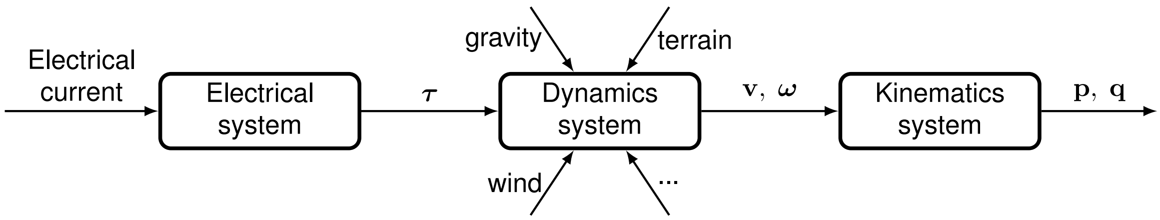
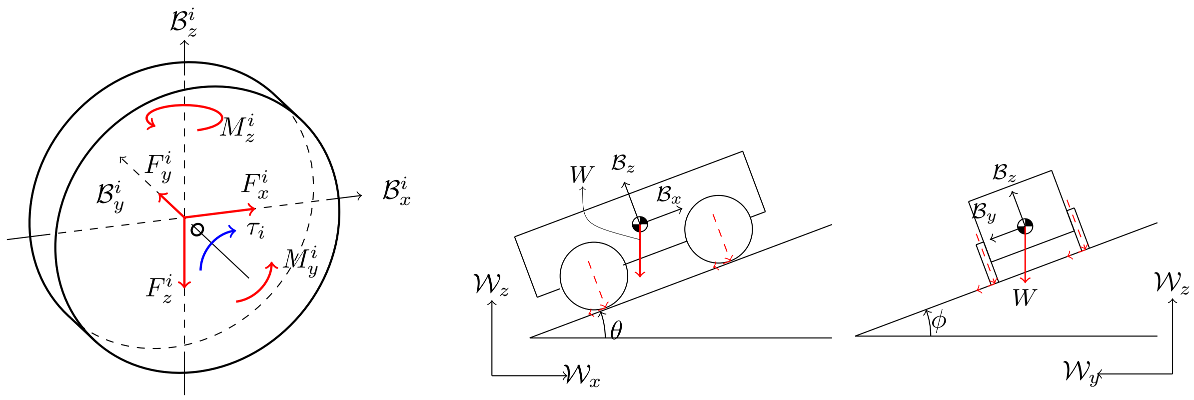


Figure 3.3: Mobile robot systems. Adapted from [1]

The presented wheeled mobile robot model contains four controllable joints at each wheel COM, which are located at each wheel's center. The mobile robot may roll, pitch or fall vertically with respect to frame $\mathcal{B}$, because it will be moving through irregular terrain, which changes the balance of forces the mobile robot is subject to. Moreover, external forces (e.g. friction contact, wheel-terrain traction) and gravity are also taken into account in this general model. Typically, wind force is also an external force which acts on the mobile robot body at outdoor environments, but its effect was not introduced in this model.

Each wheel joint can only be actuated around the wheel horizontal axis, which is parallel to $\mathcal{B}_y$. Every wheel holds a frame $\mathcal{B}_i$ parallel to $\mathcal{B}$, centered at its COM. Then, it also holds a frame $\mathcal{T}_i$ which follows up the wheel velocity vector (section 3.3.2). Apart from it, each wheel is subject to forces and moments coming from its interaction with the terrain or due to gravity (figure 3.4). Finally, these forces and moments shall be expressed with respect to frame $\mathcal{B}$ and summed up in order to feed the dynamics relationship of the mobile robot (equation 3.16).



(a) Wheel subject to forces and moments. (b) Weight distribution during pitch. (c) Weight distribution during roll.

Figure 3.4: Forces and moments distribution.

The weight distribution on the mobile robot wheels is influenced by its roll and pitch. A dynamics model is derived based on Newton's second law as well as on the angular momentum rate of change from each wheel around its respective COM. The angular momentum rate of change of the whole mobile robot around its COM is also considered.

$$\begin{aligned} m\dot{\mathbf{v}} + m\,\boldsymbol{\omega} \times \mathbf{v} &= \mathbf{f} \\ \mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) &= \mathbf{m} \end{aligned} \tag{3.16}$$

, where m stands for the mobile robot mass. Moreover, $\mathbf{I}$ denotes the inertia tensor of the mobile robot with respect to frame $\mathcal{B}$. Furthermore, $\mathbf{f} = [f_x\ f_y\ f_z]^T$ stands for the sum of forces acting on the system and $\mathbf{m} = [l\ m\ n]^T$ denotes the sum of moments. $\mathbf{f}$ and $\mathbf{m}$ are expressed with respect to frame $\mathcal{B}$.

Equation 3.16 may be obtained by applying the second newton law for translational and rotational dynamics. It is of interest to find out a dynamics formulation which is expressed with respect to $\mathcal{B}$, because $\mathbf{I}$ is constant there, but also because it is more intuitive to think of velocities, forces and moments with respect to $\mathcal{B}$. In case the linear and angular moments are defined by $\mathbf{l} = m\mathbf{v}$ and $\mathbf{h} = \mathbf{I}\boldsymbol{\omega}$ respectively, it is possible to derive a dynamics formulation with the help of equation A.7. From now on, for the sake of completeness, each moment and force acting on the mobile robot are presented.

3.4.1 Wheel actuation

Every wheel is commanded by a torque, $\boldsymbol{\tau}_i$, which is expressed with respect to $\mathcal{B}$ and it only holds a component in $\mathcal{B}_y$. That torque yields a traction force $\mathbf{t}_i$ which is expressed with respect to the wheel frame $\mathcal{T}_i$ (figure 3.4). The traction force $\mathbf{t}_i$ only holds a single longitudinal component within $\mathcal{T}^i$. $\mathbf{t}_i$ is the result of the action of the soil on the wheel at c_i. Therefore, the force produced by $\boldsymbol{\tau}_i$ on the soil opposes $\mathbf{t}_i$. The wheel frame $\mathcal{T}_i$ is centered at the wheel center and follows up the wheel velocity vector, which may not match frame $\mathcal{B}_i$. Finally, it comes that,

$$\boldsymbol{\tau}_i = -\mathbf{w}_i \times \left(\mathbf{R}^{\mathcal{B}}_{\mathcal{T}^i}\mathbf{t}_i\right),\ i = 1, ..., 4 \tag{3.17}$$

, where $\mathbf{w}_i$ stands for a vector which connects each wheel center to c_i, with respect to $\mathcal{B}$. Therefore, in order to determine the total force and moment produced by wheel actuation, one needs to express every traction force $\mathbf{t}_i$ with respect to $\mathcal{B}$, by taking into account equation 3.6. Then, the sum of traction forces, $\mathbf{t}$, with respect to $\mathcal{B}$ are expressed as,

$$\mathbf{t} = \sum_{i=1}^{4} \mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} \mathbf{t}_i \tag{3.18}$$

Assumption 5. *Moments produced around $\mathcal{B}_x$ and $\mathcal{B}_y$ are neglected because the terrain counter-acts its effects.*

The traction forces produced by wheel torque actuation may also output a moment around COM with respect to $\mathcal{B}$.

$$\mathbf{m_t} = \sum_{i=1}^{4} \mathbf{c}_i \times \left(\mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} \mathbf{t}_i\right) \tag{3.19}$$

, where $\mathbf{c}_i$ is a vector connecting the mobile robot COM to the contact point c_i, with respect to $\mathcal{B}$. Due to assumption 5, $\mathbf{m_t}$ does not include $\boldsymbol{\tau}_i$.

Finally, if the sum of moments and forces is presented with respect to the actuation $\boldsymbol{\tau}_i$, it comes that

$$\mathbf{f_t} = \sum_{i=1}^{4} \mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}} \left(\mathbf{S}^T\left(\mathbf{w}_i\right) \mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}}\right)^{\dagger} \boldsymbol{\tau}_i \tag{3.20}$$

$$\mathbf{m_t} = \sum_{i=1}^{4} \mathbf{S}\left(\mathbf{c}_i\right) \mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}} \left(\mathbf{S}^T\left(\mathbf{w}_i\right) \mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}}\right)^{\dagger} \boldsymbol{\tau}_i \tag{3.21}$$

3.4.2 Wheel spin rate

Each wheel is considered to spin only if a torque $\boldsymbol{\tau}_i$ is fed to the wheel along its lateral axis.

$$\mathbf{I}_{\text{wheel}}\, \dot{\boldsymbol{\omega}}_i = \boldsymbol{\tau}_i \tag{3.22}$$

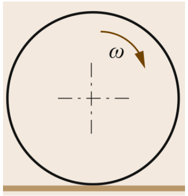
Equation 3.22 is expressed with respect to $\mathcal{B}$.

Assumption 6. *The wheel is considered to be locked, so its motion is only due to actuation control.*

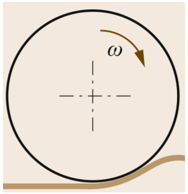
3.4.3 Wheel-terrain contact model

The ability to accurately model the interaction forces and moments between a wheel and the contact terrain is of the utmost importance, in order to simulate a theoretical model of a robot or to control a real-life application at a given environment.

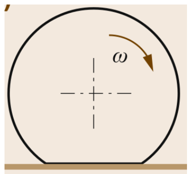
There are four possible cases of wheel-terrain interaction (figure 3.5): rigid wheel on rigid terrain, rigid wheel on deformable terrain, deformable wheel on rigid terrain and a deformable wheel on deformable terrain. In [1, 17], one can find analytical models for all these possibilities.



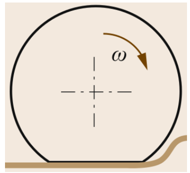
(a) Rigid wheel contacting with rigid terrain



(b) Rigid wheel contacting with deformable terrain



(c) Deformable wheel contacting with rigid terrain



(d) Deformable wheel contacting with deformable terrain

Figure 3.5: Wheel-terrain contact models [1]

In this thesis, only rigid wheels are considered. Even if a wheel presents some degree of deformability (i.e. pneumatic tire), it can be modeled as a rigid wheel if the average tire ground pressure is higher than the critical ground pressure, p_{gcr} [1].

When it comes to the type of soil, a rover may traverse deformable (sand) or rigid terrain (rock) when it is deployed at a realistic planetary environment, so it is difficult to predict *a priori* what kind of soil it will be traversing as well as its properties.

The theory of terramechanics focuses on studying the mechanical properties of soils and to assess its behaviour, when it is subject to the influence of a robot with a locomotion set. Apart from it, terramechanics models also deliver performance quantification by describing the stress fields that the contact modules are exposed to. In case of deformable soils, the standard model descriptor is the Bekker-Wong model. On the other hand, the Coulomb friction model fits well to describe the contact between rigid wheels and rigid terrain.

In this thesis, deformable terrain contact physics is discarded as a simulation option because the physics engine of Gazebo, ODE, does not provide a test-bed for it, but also due to the fact that it is cumbersome to online estimate the parameters of deformable terrain. There are open-source terramechanics real-time simulators (Project Chrono, Vortex) which could enable the simulation of deformable terrain, however these simulators do not provide a bridge to ROS, which is used as a message transmission mechanism between nodes to exchange data among multiple parallel processes. The Coulomb friction model is implemented within ODE to match as close as possible the contact physics between rigid objects.

Coulomb friction model

This model provides a friction force, $\mathbf{f_f}$, acting at each wheel, as a result of the action of the soil on the wheel. This forces acts at c_i with a direction opposite to the wheel velocity $\mathbf{v}_i$. $\mathbf{f_f}$ breaks into a longitudinal ($f_{f,x}$) and lateral ($f_{f,y}$) component, in case f_f is expressed with respect to $\mathcal{T}_i$. Apart from this, a terrain reaction force $\mathbf{f_n}$, acts at c_i, with a direction parallel to the contact normal at c_i, so $\mathbf{f_n}$ only holds a single vertical component within $\mathcal{T}^i$. The Coulomb friction model states the following,

$$\begin{cases} \left(\frac{f_{f,x}}{\mu_x f_n}\right)_i^2 + \left(\frac{f_{f,y}}{\mu_y f_n}\right)_i^2 \leq 1 \\ (f_n)_i > 0 \end{cases} \quad , \ i = 1, ..., 4 \tag{3.23}$$

Equation 3.23 states that $(\mathbf{f_f})_i$ lies within an ellipse with diagonals $\mu_x (f_n)_i$ and $\mu_y (f_n)_i$. In case $\mu_x = \mu_y$, the ellipse turns into a circle. μ_x and μ_y denote friction coefficients along the longitudinal and lateral components of $\mathcal{T}^i$, respectively.

Assumption 7. *It is assumed* $\mu_x = \mu_y$.

If a wheel i sticks to the contact point c_i, it means equation 3.23 is smaller than one. On the other hand, if a wheel is sliding, equation 3.23 equals one and $(\mathbf{f_f})_i$ has an opposite direction to $(\mathbf{v}_i)_{\mathcal{T}^i}$.

$$\begin{cases} (\mathbf{f_f})_i = -(\mathbf{t}_i + \mathbf{g}_i), \ \mathbf{v}_i = \mathbf{0} \wedge \mathbf{a}_i = \mathbf{0} \\ (\mathbf{f_f})_i = -\left\| \left(\sqrt{(\mu f_n)^2 - (f_{f,y})^2}; f_{f,y}; 0 \right) \right\| \cdot \left(\frac{\mathbf{v}_i}{\|\mathbf{v}_i\|} \right)_{\mathcal{T}^i}, \ \mathbf{v}_i \neq \mathbf{0} \\ (\mathbf{f_f})_i = \left\| \left(\sqrt{(\mu f_n)^2 - (f_{f,y})^2}; f_{f,y}; 0 \right) \right\| \cdot \left(\frac{\mathbf{a}_i}{\|\mathbf{a}_i\|} \right)_{\mathcal{T}^i}, \ \mathbf{v}_i = \mathbf{0} \wedge \mathbf{a}_i \neq \mathbf{0} \end{cases} \tag{3.24}$$

The sum of friction forces $\mathbf{f_f}$ acting on the mobile robot and the yielded sum of moments are given by,

$$\mathbf{f_f} = \sum_{i=1}^{4} \mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} (\mathbf{f_f})_i \quad \mathbf{m}_{(\mathbf{f_f})} = 0 \tag{3.25}$$

Finally, the determination of $(\mathbf{f_n})_i$ is made by computing the portion of the mobile robot weight every wheel i supports, which is presented in 3.4.3.

Effects of gravity

Gravity pushes the mobile robot into the terrain. As a response, ground reaction forces $(\mathbf{f_n})_i$ sustain the mobile robot. At each wheel i, the effect of gravity acting at c_i is denoted by $(\mathbf{f_g})_i$, which is expressed with respect to $\mathcal{T}_i$. The vertical component of $(\mathbf{f_g})_i$ equals $(\mathbf{f_n})_i$ and are orthogonal to $(\mathbf{f_f})_i$ and $\mathbf{t}_i$. The sum of $(\mathbf{f_g})_i$ for every wheel equals the mobile robot weight.

$$m\mathbf{g} = \mathbf{R}_{\mathcal{B}}^{\mathcal{W}} \sum_{i=1}^{4} \mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} (\mathbf{f_g})_i \quad , \quad (f_{g,z})_i = -(f_{n,z})_i \tag{3.26}$$

, where $\mathbf{g}$ denotes the gravity vector with respect to $\mathcal{W}$. The sum of the effects of gravity and ground reaction forces with respect to $\mathcal{B}$ are stated by,

$$\mathbf{f_g} = \sum_{i=1}^{4} \mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} (\mathbf{f_g})_i \quad \mathbf{f_n} = \sum_{i=1}^{4} \mathbf{R}_{\mathcal{T}^i}^{\mathcal{B}} (\mathbf{f_n})_i \tag{3.27}$$

$$\mathbf{m}_{(\mathbf{f_n})} = \mathbf{m}_{(\mathbf{f_g})} = \mathbf{0} \tag{3.28}$$

3.5 Full state-space model

In order to sum up the previous sections, it is possible to deliver a full state-space model with respect to the state vector $\mathbf{x} = \begin{bmatrix} \mathbf{p}^T & \mathbf{\Psi}^T & \mathbf{v}^T & \boldsymbol{\omega}^T & \boldsymbol{\omega}_i^T \end{bmatrix}^T$.

$$
\mathbf{f}(\mathbf{x}) = \begin{cases} \dot{\mathbf{p}} = \mathbf{R}_{\mathcal{B}}^{\mathcal{W}} \mathbf{v} \\ \dot{\mathbf{\Psi}} = \bar{\mathbf{R}}_{\mathcal{B}}^{\mathcal{W}} \boldsymbol{\omega} \\ \dot{\mathbf{v}} = -\boldsymbol{\omega} \times \mathbf{v} + \frac{1}{m}\mathbf{f} \\ \dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}\left(-\boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) + \mathbf{m}\right) \\ \dot{\boldsymbol{\omega}}_i = \mathbf{I}_{\text{wheel}}^{-1} \boldsymbol{\tau}_i \end{cases} \tag{3.29}
$$

Motivated from equations 3.20, 3.21, 3.25 and 3.27, $\mathbf{f}$ and $\mathbf{m}$ are represented with respect to the actuation wheel torques, $\boldsymbol{\tau}_i$.

$$
\begin{bmatrix} \mathbf{f} \\ \mathbf{m} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \mathbf{A}_3 & \mathbf{A}_4 & \mathbf{f}_\mathbf{f} & \mathbf{f}_\mathbf{g} & \mathbf{f}_\mathbf{n} \\ \mathbf{S}(\mathbf{c}_1)\mathbf{A}_1 & \mathbf{S}(\mathbf{c}_2)\mathbf{A}_2 & \mathbf{S}(\mathbf{c}_3)\mathbf{A}_3 & \mathbf{S}(\mathbf{c}_4)\mathbf{A}_4 & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}_1 \\ \boldsymbol{\tau}_2 \\ \boldsymbol{\tau}_3 \\ \boldsymbol{\tau}_4 \\ \mathbf{1} \\ \mathbf{1} \\ \mathbf{1} \end{bmatrix} \tag{3.30}
$$

, where $\mathbf{A}_i = \mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}} \left(\mathbf{S}^T(\mathbf{w}_i)\,\mathbf{R}_{\mathcal{T}_i}^{\mathcal{B}}\right)^{\dagger}$.

3.6 Model verification

Chapter 4

Background

4.1 Two-layer map generation

Elevation data is stored as a **Digital Elevation Models**, which is defined by a compact, discrete function $e_d : \mathcal{E}'_x \times \mathcal{E}'_y \rightarrow \mathcal{E}'_z$. Similarly, a **discrete traversability map** $\mathcal{T}'$ is defined by a function t_d : $\mathcal{T}'_x \times \mathcal{T}'_y \rightarrow \mathcal{T}'_z$, where $\mathcal{E}'_k$ and $\mathcal{T}'_k$ define each discrete map domain on the respective Degree of freedom. As mentioned in section 3.2.1,

$$\mathcal{E}'_x \times \mathcal{E}'_y = \mathcal{T}'_x \times \mathcal{T}'_y \tag{4.1}$$

, so t_d and e_d can be encapsulated into a **two-layer discrete map function** $m'_2 : \mathcal{M}'_{2,x} \times \mathcal{M}'_{2,y} \rightarrow \mathcal{E}'_z \times \mathcal{T}'_z$ (equation 4.2), where $\mathcal{M}'_{2,x} \times \mathcal{M}'_{2,y}$ swaps the equality introduced in 4.1 by an unique formulation.

$$(e_d(x_i, y_j), t_d(x_i, y_j)) = m_2(x_i, y_j) = (h_d, c_d), \quad i = 0, ..., N_x - 1 \;\; j = 0, ..., N_y - 1 \tag{4.2}$$

, where N_x and N_y stand as the number of divisions in the longitudinal and lateral directions of maps $\mathcal{E}'$ and $\mathcal{T}'$ along frame $\mathcal{W}$, respectively. $x_i \in \mathcal{M}'_{2,x}$ and $y_i \in \mathcal{M}'_{2,y}$, denote the longitudinal and lateral positions of the discrete two-layer map $\mathcal{M}'_2$ with respect to frame $\mathcal{W}$. $\mathcal{E}_k$, $\mathcal{T}_k$ denote continuous subsets of $\mathbb{R}$, which encapsulate each map limit on the respective Degree of freedom. Then, if functions e_d and t_d are interpolated, a **continuous elevation map function** $e : \mathcal{E}_x \times \mathcal{E}_y \rightarrow \mathcal{M}_z$ and a **continuous traversability map function** $t : \mathcal{T}_x \times \mathcal{T}_y \rightarrow \mathcal{T}_z$ can be encapsulated into a **two-layer continuous map function** $m_2 : \mathcal{M}_{2,x} \times \mathcal{M}_{2,y} \rightarrow \mathcal{E}_z \times \mathcal{T}_z$,

$$(e(x, y), t(x, y)) = m_2(x, y) = (h, c), \quad x \in \mathcal{M}_{2,x}, \quad y \in \mathcal{M}_{2,y} \tag{4.3}$$

A discrete or continuous **traversability map** can be obtained from e_d or e through an algorithm $\mathbb{T}$, which is presented in section insert section; draw picture of 2-layer map.

4.2 Fast Marching Method

In this thesis, a fast marching method is employed in order to serve as the framework to extract optimal reference paths. This method relies on solving a boundary-value problem [18], which yields a potential field with a wave that starts at a pre-defined source and expands throughout a map $\mathcal{M}_1 : \mathbb{R}^2 \longmapsto \mathbb{R}$, which is a reduction of a layered map $\mathcal{M}_n : \mathbb{R}^2 \longmapsto \mathbb{R}^n$.

Mathematically, this comes to solving the Eikonal equation,

$$\begin{aligned} |\nabla u(x,y)|\, f(x,y) = 1 \\ f(\Gamma) = 0 \end{aligned} \quad , \qquad (x,y),\ \Gamma \in D_{\mathcal{M}_n} \tag{4.4}$$

, where $u(.)$ is the solution to the Eikonal equation, (x,y) is a position in the map, $f(.)$ is a speed (or cost) function [10] which contains information regarding the set of points where the wave delays or rushes its progression throughout $\mathcal{M}_2$. $\Gamma \subset \mathcal{M}_2$ is the wave front at the source. Due to the fact that $u(\boldsymbol{x})$ delivers the time a wave needs to reach $\boldsymbol{x}$ from the source $\boldsymbol{x}_s \in \mathcal{M}_2$, it is trivial that $u(\Gamma)$ equals zero because Γ is small ball around $\boldsymbol{x}_s$. On the other hand, $u(\boldsymbol{x}) = T$ provides the set of points from where it takes a time T to reach $\boldsymbol{x}_s$.

Then, it is possible to find an optimal time path from $\boldsymbol{x}$ to $\boldsymbol{x}_s$ which deviates from the points where F has the lowest values. The direction of a path at some point is perpendicular to the wave front at that point, which is provided by the gradient $-\Delta u(\boldsymbol{x})$. The function F is computed offline according to the application, and then it is inserted into 4.4. After that, the eikonal problem is solved (also offline) and delivers a potential field with a source $\boldsymbol{x}_s$. From there, it is possible to compute online the fastest and safest path to $\boldsymbol{x}_s$ at the position the mobile robot is in. More information on this issue is provided at

Chapter 5

Global path planner

In order to feed the control system with a reference path, an offline survey of the target environment is conducted in order to generate a **traversability cost map**, which performs an analysis of the terrain uneveness with respect to the robot kinematics and geometric features. Then, the same **cost map** is fed to the Eikonal equation in order to yield a gradient potential map from where an optimal path to a pre-determined goal is retrieved.

5.1 Traversability map computation

As mentioned in subsection 2.1.1, there is no explicit guideline on how to generate a traversability map. In this thesis, several approaches are compared. First, the state-of-the art algorithm from [7] is reviewed and evaluated. Then, this algorithm is reformulated to account for other metrics such as terrain inclination and roughness [9].

5.1.1 Height minimization

A constrained optimization problem which minimizes the height of the robot at each cell is described. It is very similar to the solution provided in [7]. In this case, a wheeled vehicle is used instead of a tracked one. Moreover, the yaw angle is set as a free variable and the initial guess $\boldsymbol{v}_0$ of the problem at every cell is set by an approximation to the optimal solution.

Initial guess

For a range of yaws between $-\pi$ and π, each wheel point is projected to $\mathcal{M}_2$. The centroid of the projected points is computed, and the yaw and contact points of the lowest height centroid are selected. Then, from the group of retrieved contact points, the lowest height one is discarded, and an unit, normal vector to a plane which contains these points, is computed. Finally, the roll and pitch of this plane are given by equation 5.1, where $(a,\, b,\, c)$ are the coordinates of a normal vector to the above mentioned plane. In case any of these angles are higher than ϕ_{max} or θ_{max}, the respective angle is set to the highest admissable value, so that the initial guess is feasible. The height initial guess z_0 is set as the sum of the projection of the robot center of mass point to $\mathcal{M}_2$ plus the robot geometric height, to avoid any possible intrusion of the vehicle hull into the ground.

$$\begin{aligned} \phi &= \arcsin\left(\sin(\psi)\, a - \cos(\psi)\, b\right) \\ \theta &= \arccos\left(\frac{c}{\cos(\phi)}\right) \end{aligned} \tag{5.1}$$

Non-linear programming

The minimization of the vehicle Center of mass is expressed as a non-linear optimization problem (NLP) in equation 5.2, where $\boldsymbol{\Psi}_{max}$ stands as the limits for each euler angle. $\mathbf{p}_k$ stand as the k hull point position, which is defined with respect to $\boldsymbol{p}$ and $\boldsymbol{\Psi}$, by taking into account the robot geometry. $[x, y]^T$ are fixed variables whereas $[z,\, \phi,\, \theta,\, \psi]^T$ are optimization variables. Each point included in 5.2 adds an additional constraint by stating each hull point must be above or at contact with the terrain. In case less than three hull points are at contact with the terrain, the terrain inclination returned by algoritm 1 is $\frac{\pi}{2}$. This expresses the fact that the robot will stand at an unstable position, so this position is regarded as an undesirable traversable position. On the other hand, the terrain inclination is given by equation 5.3, which returns the angle between a normalized vector with direction $\hat{\mathcal{B}}_z$ and the normalized vector $\mathcal{W}_z$ [7]. Although algorithm 1 can be extended to any kind of mobile robot configuration, only four wheel mobile robots are considered in this thesis, as mentioned in section 3.1.

$$\begin{aligned} \text{Minimize} \quad & J = z \\ \text{w.r.t.} \quad & \boldsymbol{v} = (z,\, \phi,\, \theta,\, \psi) \\ \text{subject to} \quad & \forall_k : z_k - e(x_k, y_k) \geq 0 \\ & \boldsymbol{\Psi}_{max} - |\boldsymbol{\Psi}| \geq 0 \end{aligned} \tag{5.2}$$

$$c = \arccos(\cos(\theta)\cos(\phi)) \tag{5.3}$$

Algorithm 1 Terrain inclination with HEIGHTMIN

function TERRAINSLOPE($\boldsymbol{\Psi}_{max},\, e(\cdot),\, \boldsymbol{v}_0,\, x,\, y$)
  $\boldsymbol{v}_* \leftarrow$ HEIGHTMIN($\boldsymbol{v}_0,\, \boldsymbol{\Psi}_{max},\, e(\cdot),\, x,\, y$) ▷ solve 5.2
  if Contacts ≥ 3 **then**
    $c \leftarrow$ INCLINATION($\phi_*,\, \theta_*$) ▷ 5.3
  else
    $c \leftarrow \frac{\pi}{2}$
  end if
  return $c,\, \boldsymbol{v}_*$
end function

Then, algorithm 2 extends the application of algorithm 1 to each cell of the elevation grid. It should be noted that the discrete elevation grid $e_d(\cdot)$ shall have such dimensions that the cell divison length is of the same magnitude order of the wheel diameter in meters. Moreover, a convex area $\mathcal{C}_2$ (equation 5.4) centered at $[x,\, y]^T$ which captures the robot hull $\mathcal{H}_2$ projection in $\mathcal{M}_2$ must lie inside the map domain. If this condition is not met, the terrain inclination angle assigned to $t_d(\cdot)$ is $\frac{\pi}{2}$. Therefore, positions at which the

Algorithm 2 Terrain inclination map with TERRAINSLOPE

Require: $\boldsymbol{\Psi}_{max},\, e(\cdot),\, N_0,\, N_1$
for each $\{i,\, j\} \in \{0, \cdots, N_0 - 1\} \times \{0, \cdots, N_1 - 1\}$ **do**
  $\boldsymbol{v}_0 \leftarrow$ INITIALGUESS($i,\, j$) ▷ feasible initialization
  if $\mathcal{C}_2 \subset \mathcal{M}_2$ **then**
    $c,\, \boldsymbol{v}_* \leftarrow$ TERRAINSLOPE($\boldsymbol{\Psi}_{max},\, e(\cdot),\, \boldsymbol{v}_0, x_i,\, y_j$) ▷ algorithm 1
  else
    $c \leftarrow \frac{\pi}{2}$
  end if
  $t_d(i,\, j) \leftarrow c$
end for

assigned terrain inclination is $\frac{\pi}{2}$ are considered to be untraversable. To sum up, the higher the inclination, the less desirable it is to pass through the respective position.

$$\mathcal{C}_2 = \{\text{min}\, \underline{h} : \|\mathbf{p}_2 - \mathbf{a}\| \leq \underline{h} \,|\, \mathbf{p}_2,\, \mathbf{a} \in \mathcal{M}_2,\, \mathbf{p}_{2,k} \in \mathcal{C}_2\} \tag{5.4}$$

5.1.2 Close to ground

A slight reformulation of the height minimization problem is presented in equation 5.5, so that an optimal solution matches with a small objective function value. The initial guess determination remains exactly the same. The terrain inclination map generation also remains equal to the one presented in algorithm 2.

$$\begin{aligned} \text{Minimize} \quad & J = (z - e(x,\, y))^2 \\ \text{w.r.t.} \quad & \boldsymbol{\upsilon} = (z,\, \phi,\, \theta,\, \psi) \\ \text{subject to} \quad & \forall_k : z_k - e(x_k, y_k) \geq 0 \\ & \boldsymbol{\Psi}_{max} - |\boldsymbol{\Psi}| \geq 0 \end{aligned} \tag{5.5}$$

5.1.3 Maximum terrain roughness

Terrain traversability may also be evaluated by inspecting terrain roughness around each position of the map. The goal is evaluate the maximum terrain uneveness at each position of the map, in order to generate a worst-case scenario cost map. In this thesis, terrain roughness at each position of $e_d(\cdot)$ is defined as the standard deviation of elevations inside $\mathcal{H}_2$.

$$\bar{e}_d(i, j) = \frac{\sum_{x_i,\, y_j \in \mathcal{H}_2} e(x_i,\, y_j)}{n(\underline{\mathcal{H}}_2)},\ \underline{\mathcal{H}}_2 = \{(x_i,\, y_j) \in \mathcal{H}_2\} \tag{5.6}$$

, where $\underline{\mathcal{H}}_2$ denotes a finite set of points from $e_d(\cdot)$ inside $\mathcal{H}_2$ and $\bar{e}_d(i, j)$ the average of elevations for points inside $\mathcal{H}_2$. Then, terrain roughness is defined as,

$$t_d(i,\, j) = \sqrt{\frac{\sum_{x_i,\, y_j \in \mathcal{H}_2} \left(e(i,\, j) - \bar{e}(i,\, j)\right)^2}{n(\underline{\mathcal{H}}_2)}} \tag{5.7}$$

However, a NLP only accepts continuous functions, so the above equations turn into,

$$\bar{e}(x,\, y) = \frac{\int\int_{\mathcal{H}_2} e(x_i,\, y_j)\, dx\, dy}{\mathcal{A}(\mathcal{H}_2)} \tag{5.8}$$

$$t(x,\, y) = \sqrt{\frac{\int\int_{\mathcal{H}_2} \left(e(x_i,\, y_j) - \bar{e}(x,\, y)\right)^2\, dx\, dy}{\mathcal{A}(\mathcal{H}_2)}} \tag{5.9}$$

, where $\mathcal{A}(\mathcal{H}_2) = \int\int_{\mathcal{H}_2} 1\, dx\, dy$.

Initial guess

The determination of an initial guess is made in a similar way to what was presented in 5.1.1. However, the initial guess determination method is reformulated in order to find the pose with maximum roughness. For every yaw, the robot hull points are projected to $\mathcal{M}_2$ and the roughness for the section of terrain inside that area is computed (equation 5.9). Then, the maximum roughness yaw is retrieved. At the end, the roll, pitch and height are determined as it is described in 5.1.1.

Non-linear programming

Due to that the fact this optimization problem implies a maximization, the objective function $J(\cdot)$ must be manipulated in order to mirror that approach. Therefore, a cost function of $\exp(-t(x,\, y))$ is selected because its minimum matches the maximum of $t(x,\, y)$. On the height minimization problem, the vehicle is naturally pushed to the ground, which is not the case this time. Then, the wheels contact to the ground are set as hard constraints, so that a maximum roughness solution is one where the vehicle wheels are at contact with the ground.

Assumption 8. *A wheel is at contact with the ground if the distance between the lowest height wheel point and the ground is* $r \times 10^{-2}$.

$$\begin{aligned}
\text{Minimize} \quad & J = \exp(-t(x,\, y)) \\
\text{w.r.t.} \quad & \boldsymbol{\upsilon} = (z,\, \phi,\, \theta,\, \psi) \\
\text{subject to} \quad & z - e(x,y) \geq 0 \\
& \forall_k : z_k - e(x_k, y_k) \leq r \times 10^{-2} \\
& \boldsymbol{\Psi}_{max} - |\boldsymbol{\Psi}| \geq 0
\end{aligned} \tag{5.10}$$

In case $\mathcal{C}_2 \subset \mathcal{M}_2$, the roughness is saved as `None`. Later on, a decimal value will be attributed to the respective cell. The computation of the area integrals from equations 5.8 and 5.9 are obtained by application of a Gauss-Legendre quadrature (A.7) to each integral separately. The integrals limits are selected by finding the wheel points with the minimum and maximum x and y coordinates, by using the Boltzmann operator (A.8), which is a smooth function that can be inserted into a NLP, that computes the maximum and minimum among an array of values.

5.1.4 Maximum inclination

A last option to evaluate the terrain traversability is to find the maximum inclination at each position of $e(\cdot)$. The goal is to generate a worst-case scenario cost map at each position, so that the global planner is aware of the maximum risk at each position.

Initial guess

The initial guess determination method is reformulated (once again), so that the highest inclination approximated pose is determined. However, the method core remains the same as the one in 5.1.1.

Non-linear programming

The vehicle inclination angle is given by equation 5.3, so the maximum by inclination angle may be found by stating the cost function $J(\cdot)$ as $\cos(\phi)\cos(\theta)$. Once again, it needs to be set as hard constraint the fact that the vehicle wheels must be at contact with the ground, so that a maximum inclination solution at each cell is one where the wheels touch the ground.

$$\begin{aligned} \text{Minimize} \quad & J = \cos(\phi)\cos(\theta) \\ \text{w.r.t.} \quad & \boldsymbol{v} = (z,\, \phi,\, \theta,\, \psi) \\ \text{subject to} \quad & z - e(x, y) \geq 0 \\ & \forall_k : z_k - e(x_k, y_k) \leq r \times 10^{-2} \\ & \boldsymbol{\Psi}_{max} - |\boldsymbol{\Psi}| \geq 0 \end{aligned} \tag{5.11}$$

Apart from the NLP (equation 5.11), algorithm 2 is also applied to the maximum inclination problem.

5.1.5 Implementation

In this subsection, the implentation of the NLP's presented above is described, alongside the presentation of results and its discussion. The elevation map used in this implementation is the target environment, which will serve as a simulation test-bed for the control system, later on. The wheeled mobile robot Pioneer 3AT was chosen as the test vehicle. The elevation map and mobile robot dimensions are presented in table 5.1 and **??**.

Table 5.1: Pioneer 3-AT dimensions.

r [m]	l_w [m]	b_w [m]
0.111	0.268	0.394

The selected elevation map is presented below,

5.2 Fast Marching

Up to now, it is possible to compute a potential flow based on the cost introduced in equation **??**. This cost replaces function $F(\cdot)$ in equation 4.4 and the solution of Eikonal problem outputs a gradient which enable the computation of an optimal path towards the selected goal.

After the discretization of gradient ∇u, the optimal path comes as,

$$\alpha = \arctan \frac{\frac{\partial}{\partial y} u(p_i)}{\frac{\partial}{\partial x} u(p_i)} \tag{5.12}$$

$$p_{i+1,\, x} = p_{i,\, x} - cos(\alpha) \tag{5.13}$$

$$p_{i+1,\, y} = p_{i,\, y} - sin(\alpha) \tag{5.14}$$

Chapter 6

Results

Insert your chapter material here...

6.1 Problem Description

Description of the baseline problem...

6.2 Baseline Solution

Analysis of the baseline solution...

6.3 Enhanced Solution

Quest for the optimal solution...

6.3.1 Figures

Insert your section material and possibly a few figures...

Make sure all figures presented are referenced in the text!

The caption should appear below the figure.

6.3.1.1 Images

Figure 6.1: Caption for figure.

(a) Airbus A320 (b) Bombardier CRJ200

Figure 6.2: Some aircrafts.

Make reference to Figures 6.1 and 6.2.

By default, the supported file types are *.png,.pdf,.jpg,.mps,.jpeg,.PNG,.PDF,.JPG,.JPEG*.

See `http://mactex-wiki.tug.org/wiki/index.php/Graphics_inclusion` for adding support to other extensions.

6.3.1.2 Drawings

Insert your subsection material and for instance a few drawings...

The schematic illustrated in Fig.6.3 can represent some sort of algorithm.

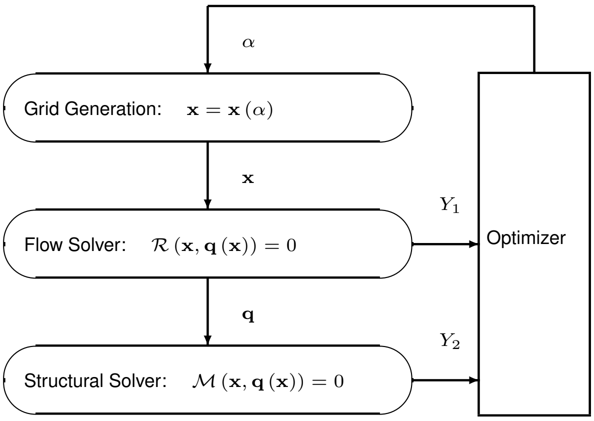


Figure 6.3: Schematic of some algorithm.

6.3.2 Equations

Equations can be inserted in different ways.

The simplest way is in a separate line like this

$$\frac{\mathrm{d}q_{ijk}}{\mathrm{d}t} + \mathcal{R}_{ijk}(\mathbf{q}) = 0\,. \tag{6.1}$$

If the equation is to be embedded in the text. One can do it like this $\partial\mathcal{R}/\partial\mathbf{q} = 0$.

It may also be split in different lines like this

$$\begin{aligned} \text{Minimize} \quad & Y(\alpha, \mathbf{q}(\alpha)) \\ \text{w.r.t.} \quad & \alpha\,, \\ \text{subject to} \quad & \mathcal{R}(\alpha, \mathbf{q}(\alpha)) = 0 \\ & C(\alpha, \mathbf{q}(\alpha)) = 0\,. \end{aligned} \tag{6.2}$$

It is also possible to use subequations. Equations (6.3a), (6.3b) and (6.3c) form the Naver–Stokes equations (6.3).

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho u_j\right) = 0\,, \tag{6.3a}$$

$$\frac{\partial}{\partial t}\left(\rho u_i\right) + \frac{\partial}{\partial x_j}\left(\rho u_i u_j + p\delta_{ij} - \tau_{ji}\right) = 0, \quad i = 1, 2, 3\,, \tag{6.3b}$$

$$\frac{\partial}{\partial t}\left(\rho E\right) + \frac{\partial}{\partial x_j}\left(\rho E u_j + p u_j - u_i \tau_{ij} + q_j\right) = 0\,. \tag{6.3c}$$

6.3.3 Tables

Insert your subsection material and for instance a few tables...

Make sure all tables presented are referenced in the text!

The caption should appear above the table.

Follow some guidelines when making tables:

- Avoid vertical lines
- Avoid “boxing up” cells, usually 3 horizontal lines are enough: above, below, and after heading
- Avoid double horizontal lines
- Add enough space between rows

Table 6.1: Table caption.

Model	C_L	C_D	C_{My}
Euler	0.083	0.021	-0.110
Navier–Stokes	0.078	0.023	-0.101

Make reference to Tab.6.1.

Tables 6.2 and 6.3 are examples of tables with merging columns:

An example with merging rows can be seen in Tab.6.4.

If the table has too many columns, it can be scaled to fit the text widht, as in Tab.6.5.

Table 6.2: Memory usage comparison (in MB).

	Virtual memory [MB]	
	Euler	Navier–Stokes
Wing only	1,000	2,000
Aircraft	5,000	10,000
(ratio)	$5.0\times$	$5.0\times$

Table 6.3: Another table caption.

	$w = 2$			$w = 4$		
	$t = 0$	$t = 1$	$t = 2$	$t = 0$	$t = 1$	$t = 2$
$dir = 1$						
c	0.07	0.16	0.29	0.36	0.71	3.18
c	-0.86	50.04	5.93	-9.07	29.09	46.21
c	14.27	-50.96	-14.27	12.22	-63.54	-381.09
$dir = 0$						
c	0.03	1.24	0.21	0.35	-0.27	2.14
c	-17.90	-37.11	8.85	-30.73	-9.59	-3.00
c	105.55	23.11	-94.73	100.24	41.27	-25.73

Table 6.4: Yet another table caption.

ABC	header			
	1.1	2.2	3.3	4.4
IJK	group		0.5	0.6
			0.7	1.2

Table 6.5: Very wide table.

Variable	a	b	c	d	e	f	g	h	i	j
Test 1	10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000	100,000
Test 2	20,000	40,000	60,000	80,000	100,000	120,000	140,000	160,000	180,000	200,000

6.3.4 Mixing

If necessary, a figure and a table can be put side-by-side as in Fig.6.4

Legend		
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

Figure 6.4: Figure and table side-by-side.

Chapter 7

Conclusions

Insert your chapter material here...

7.1 Achievements

The major achievements of the present work...

7.2 Future Work

A few ideas for future work...

Bibliography

[1] K. Yoshida, B. Wilcox, et al. *Handbook of Robotics*. Springer Nature, second edition, 2016.

[2] W. Burgard, M. Hebert, and M. Bennewitz. *World Modeling*. 2016.

[3] P. Fankhauser, M. Bloesch, C. Gehring, M. Hutter, and R. Siegwart. Robot-centric elevation mapping with uncertainty estimates. In *Mobile Service Robotics*, pages 433–440. World Scientific, 2014.

[4] R. Correal and G. Pajares. Onboard autonomous navigation architecture for a planetary surface exploration rover and functional validation using open-source tools. In *ESA Intl. Conf. on Advanced Space Technologies in Robotics and Automation*.

[5] D. E. Smith, M. T. Zuber, H. V. Frey, J. B. Garvin, J. W. Head, D. O. Muhleman, G. H. Pettengill, R. J. Phillips, S. C. Solomon, H. J. Zwally, et al. Mars orbiter laser altimeter: Experiment summary after the first year of global mapping of mars. *Journal of Geophysical Research: Planets*, 2001.

[6] P. G. Jayasekara, G. Ishigami, and T. Kubota. Testing and validation of autonomous navigation for a planetary exploration rover using opensource simulation tools. In *International Symposium on Artificial Intelligence, Robotics and Automation in Space*.

[7] D. Amorim and R. Ventura. Towards efficient path planning of a mobile robot on rough terrain. In *2014 ieee international conference on autonomous robot systems and competitions (icarsc)*.

[8] Y. Yokokohji, S. Chaen, and T. Yoshikawa. Evaluation of traversability of wheeled mobile robots on uneven terrains by a fractal terrain model. *Advanced Robotics*, 2007.

[9] G. Ishigami, K. Nagatani, and K. Yoshida. Path planning for planetary exploration rovers and its evaluation based on wheel slip dynamics. In *Proceedings 2007 IEEE International Conference on Robotics and Automation*.

[10] R. Ventura and A. Ahmad. Towards optimal robot navigation in domestic spaces. *Lecture Notes in Artificial Intelligence (Subseries of Lecture Notes in Computer Science)*, 8992:318–331, 1 2015.

[11] C. A. Brooks and K. Iagnemma. Self-supervised terrain classification for planetary surface exploration rovers. *Journal of Field Robotics*, 2012.

[12] P. Fankhauser and M. Hutter. A Universal Grid Map Library: Implementation and Use Case for Rough Terrain Navigation. In A. Koubaa, editor, *Robot Operating System (ROS) – The Complete Reference (Volume 1)*, chapter 5. Springer, 2016. ISBN 978-3-319-26052-5. doi: 10.1007/978-3-319-26054-9{_}5. URL http://www.springer.com/de/book/9783319260525.

[13] T. Foote and M. Purvis. Standard units of measure and coordinate conventions, 10 2010. URL https://ros.org/reps/rep-0103.html#coordinate-frame-conventions.

[14] K. Kozlowski and D. Pazderski. Modeling and control of a 4-wheel skid-steering mobile robot. *International Journal of Applied Mathematics and Computer Science*, 14:477–496, 2004.

[15] K. Iagnemma, C. Brooks, and S. Dubowsky. Visual, tactile, and vibration-based terrain analysis for planetary rovers. volume 2, pages 841–848, 2004.

[16] A. Peiret, E. Karpman, et al. Modelling of off-road wheeled vehicles for real-time dynamic simulation. *Journal of terramechanics*, pages 45–58, 4 2021.

[17] G. Genta. *Introduction to the Mechanics of Space Robots*. Springer, 2012.

[18] J. A. Sethian. Fast marching methods. *SIAM Review*, 41:199–235, 1999.

[19] C. M. Roithmayr and D. H. Hodges. *Dynamics: Theory and Application of Kane's Method*. Cambridge University Press, 2015.

[20] W. F. Phillips, C. E. Hailey, and G. A. Gebert. Review of attitude representations used for aircraft kinematics. *Journal of aircraft*, 2001.

[21] A. B. Younes, D. Mortari, et al. Attitude error kinematics. *Journal of guidance, control, and dynamics*, 2014.

Appendix A

Annex A

A.1 Euler angles

Euler rotation angles defining an intrinsic $z-y-x$ (yaw-pitch-roll) rotation, with a rotation sequence of $z-y'-x''$ around the moving frame $\mathcal{B}$ is denoted as,

$$\mathbf{R}_{\mathcal{B}}^{\mathcal{W}} = \boldsymbol{R}_z(\psi)\boldsymbol{R}_y(\theta)\boldsymbol{R}_x(\phi) \tag{A.1}$$

, with the previous multiplication to yield a mapping from frame $\mathcal{B}$ to $\mathcal{W}$, which stands as,

$$\begin{aligned}\mathbf{R}_{\mathcal{B}}^{\mathcal{W}}(\phi,\theta,\psi) &= \begin{bmatrix}\cos(\psi) & -\sin(\psi) & 0\\ \sin(\psi) & \cos(\psi) & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos(\theta) & 0 & \sin(\theta)\\ 0 & 1 & 0\\ -\sin(\theta) & 0 & \cos(\theta)\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & \cos(\phi) & -\sin(\phi)\\ 0 & \sin(\phi) & \cos(\phi)\end{bmatrix}\\ &= \begin{bmatrix}\cos(\psi)\cos(\theta) & \cos(\psi)\sin(\theta)\sin(\phi)-\sin(\psi)\cos(\phi) & \cos(\psi)\sin(\theta)\cos(\phi)+\sin(\psi)\sin(\phi)\\ \sin(\psi)\cos(\theta) & \sin(\psi)\sin(\theta)\sin(\phi)+\cos(\psi)\cos(\phi) & \sin(\psi)\sin(\theta)\cos(\phi)-\cos(\psi)\sin(\phi)\\ -\sin(\theta) & \cos(\theta)\sin(\phi) & \cos(\theta)\cos(\phi)\end{bmatrix}\end{aligned} \tag{A.2}$$

Rotation matrices are singular ($\det(\boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}}) = 1$) and symmetric, $(\boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}})^T = (\boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}})^{-1} = \boldsymbol{R}_{\mathcal{W}}^{\mathcal{B}}$. Euler rotation matrices can be derived from 24 different conventions ranging from extrinsic or intrinsic rotation with different rotation orders. More information on this can be found in [19].

A.2 Skew-symmetric matrix

A skew-symmetric matrix $\boldsymbol{S}$ is a square matrix whose transpose equals its negative, $\boldsymbol{S}^T = -\boldsymbol{S}$. In this thesis, three by three matrices are employed which take the following form,

$$\boldsymbol{S}(\boldsymbol{r} = (x,\ y,\ z)) = \begin{bmatrix}0 & -z & y\\ z & 0 & -x\\ -y & x & 0\end{bmatrix} \tag{A.3}$$

In case this matrix is multiplied by a column vector $\mathbf{a}$, it is the same as applying the cross product between $\mathbf{r}$ and $\mathbf{a}$. Then, it is stated that,

$$\mathbf{S}(\mathbf{r})\mathbf{a} = \mathbf{r} \times \mathbf{a} \tag{A.4}$$

A.3 Inertia tensor

The inertia tensor of a rigid body is given by,

$$\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} \tag{A.5}$$

, where the products of inertia are equal between each other ($I_{xy} = I_{yx}$, $I_{xz} = I_{zx}$, $I_{yz} = I_{zy}$).

A.4 Differentiation of a vector

If a vector is expressed with respect to a non-inertial frame $\mathcal{P}$, it is still possible to differentiate that same vector with respect to an inertial frame $\mathcal{Q}$ [**?**].

Let $\mathbf{p} = p_1\ \mathbf{i} + p_2\ \mathbf{j} + p_3\ \mathbf{k}$, with $\mathbf{i},\ \mathbf{j},\ \mathbf{k}$ to be the canonical vectors of a non-inertial frame $\mathcal{P}$. Then, the differentiation of $\mathbf{p}$ with respect to $\mathcal{Q}$ is given by,

$$\left.\frac{d\mathbf{p}}{dt}\right|_{\mathcal{Q}} = \dot{p}_1\ \mathbf{i} + \dot{p}_2\ \mathbf{j} + \dot{p}_3\ \mathbf{k} + p_1\frac{d\mathbf{i}}{dt} + p_2\frac{d\mathbf{j}}{dt} + p_3\frac{d\mathbf{k}}{dt} \tag{A.6}$$

, where $\left.\frac{d\mathbf{p}}{dt}\right|_{\mathcal{P}} = \dot{p}_1\ \mathbf{i} + \dot{p}_2\ \mathbf{j} + \dot{p}_3\ \mathbf{k}$ because the canonical vectors $\mathbf{i},\ \mathbf{j},\ \mathbf{k}$ do not change with respect to time if the observer is located at the origin of $\mathcal{P}$. According to the general expression $\frac{d\mathbf{r}}{dt} = \mathbf{\Omega} \times \mathbf{r}$, it comes that $\frac{d\mathbf{i}}{dt} = \mathbf{\Omega} \times \mathbf{i},\ \frac{d\mathbf{j}}{dt} = \mathbf{\Omega} \times \mathbf{j},\ \frac{d\mathbf{k}}{dt} = \mathbf{\Omega} \times \mathbf{k}$. Then, equation A.6 may be rewritten as,

$$\left.\frac{d\mathbf{p}}{dt}\right|_{\mathcal{Q}} = \left.\frac{d\mathbf{p}}{dt}\right|_{\mathcal{P}} + \mathbf{\Omega} \times \mathbf{p} \tag{A.7}$$

, where $\mathbf{\Omega}$ denotes the angular velocity of frame $\mathcal{P}$ with respect to frame $\mathcal{Q}$.

A.5 Parallel-axis theorem

A.6 Quaternions

A.6.1 Rotation matrix

A quaternion is defined by $\mathbf{q} = q_w + q_x\mathbf{i} + q_y\mathbf{j} + q_z\mathbf{k}$. The right-hand rule convention is adopted.

The rotation matrix [20] which maps linear velocities from a non-inertial frame $\mathcal{P}$ to an inertial frame $\mathcal{Q}$ is the following,

$$\bar{\mathbf{R}}_{\mathcal{P}}^{\mathcal{Q}} = \begin{bmatrix} 1-2(q_y^2+q_z^2) & 2(q_xq_y-q_zq_w) & 2(q_xq_z+q_yq_w) \\ 2(q_xq_y+q_zq_w) & 1-2(q_x^2+q_z^2) & 2(q_yq_z-q_xq_w) \\ 2(q_xq_z-q_yq_w) & 2(q_yq_z+q_xq_w) & 1-2(q_x^2+q_y^2) \end{bmatrix} \tag{A.8}$$

The rotation matrix which maps angular velocities to quaternion rates is the following,

$$\bar{\mathbf{R}}_{\mathcal{P}}^{\mathcal{Q}} = \frac{1}{2} \times \begin{bmatrix} -q_x & -q_y & -q_z \\ q_w & -q_z & q_y \\ q_z & q_w & -q_x \\ -q_y & q_x & q_w \end{bmatrix} \tag{A.9}$$

A.6.2 Error matrix

The difference between two quaternions is expressed in A.10. More information on the derivation of this result can be found in [21].

$$\delta\mathbf{q} = \mathbf{q} \otimes \mathbf{q}_{ref}^{-1} \begin{bmatrix} q_wq_w^{ref} + q_xq_x^{ref} + q_yq_y^{ref} + q_zq_z^{ref} \\ -q_wq_x^{ref} + q_xq_w^{ref} - q_yq_z^{ref} + q_zq_y^{ref} \\ -q_wq_y^{ref} + q_xq_z^{ref} + q_yq_w^{ref} - q_zq_x^{ref} \\ -q_wq_z^{ref} - q_xq_y^{ref} + q_yq_x^{ref} + q_zq_w^{ref} \end{bmatrix} \tag{A.10}$$

A.7 Gauss-Legrendre quadrature

A Gauss-Legrendre quadrature is an integral approximation method, which yields an exact result for polynomials of degree $2n-1$ or less.

$$\int_a^b f(x)\,dx \approx \int_{-1}^1 f\left(\frac{b-a}{2}\xi + \frac{a+b}{2}\right)\frac{dx}{d\xi}d\xi \tag{A.11}$$

, where $\frac{dx}{d\xi} = \frac{b-a}{2}$ expresses a change of variable in order to yield an integration in $[-1,\,1]$

Then, equation A.11 may be approximated by,

$$\int_a^b f(x)dx \approx \frac{a-b}{2}\sum_{i=1}^n w_i f\left(\frac{b-a}{2}\xi_i + \frac{a+b}{2}\right) \tag{A.12}$$

In the case of a double integral, it comes that,

$$\int_c^d \int_a^b f\,(x,\,y)\,dx\,dy \approx \frac{(a-b)(d-c)}{4}\sum_{j=1}^m \sum_{i=1}^n f\left(\frac{b-a}{2}\xi_i + \frac{a+b}{2},\, \frac{d-c}{2}\xi_j + \frac{c+d}{2}\right) \tag{A.13}$$

In this thesis, a 5-point Gauss Legrendre quadrature was used, where $\xi_i = \{-0.90618,\, -0.538469,\, 0.0,\, 0.538469,\, 0.90618\}$ and $w_i = \{0.236927,\, 0.478629,\, 0.568889,\, 0.478629,\, 0.236927\}$.

A.8 Boltzman operator

The boltzmann operator is a smooth, differentiable function that approximates the non-smooth maximum or minimum functions.

$$\mathcal{S}_\alpha(x_1, \ldots, x_n) = \frac{\sum_{i=1}^{n} x_i \exp(\alpha\, x_i)}{\sum_{i=1}^{n} \exp(\alpha\, x_i)} \tag{A.14}$$

The boltzmann operator has the following properties:

1. $\lim_{\alpha \to \infty} \mathcal{S}_\alpha = \text{max}(x_1, \ldots, x_n)$
2. $\mathcal{S}_0$ is the arithmetic mean of $(x_1, \ldots, x_n)$.
3. $\lim_{\alpha \to -\infty} \mathcal{S}_\alpha = \text{min}(x_1, \ldots, x_n)$

Appendix B

Technical Datasheets

It is possible to add PDF files to the document, such as technical sheets of some equipment used in the work.